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Sovereign Cognitive Robotics Framework

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PROJECT HNL-INTL-2026-KAZ-001

Sovereign Edge-Cognitive Robotics Architecture for Trans-Jurisdictional Autonomous Systems

Abstract: This whitepaper discloses the technical architecture, governance framework, and compliance structure for PROJECT

HNL-INTL-2026-KAZ-001, a sovereign cognitive robotics platform designated for deployment across Indian and Central Asian operational theaters. The system integrates EHARS 35.5 S-MAX high-fidelity edge

cognitive loop, FFET Power Architecture for energy sovereignty, and a 43-DOF kinematic assembly under a proprietary guardianship model. We present novel solutions to Research Gaps 1 and 2 concerning sub-millisecond cognitive latency and perpetual energy operation. The framework achieves regulatory compliance with India IT Intermediary Rules 2026, UK Property Act 2025, and Pax Silica Declaration 2026 while maintaining enforcement immunity through jurisdictional shielding. Commercial deployment targets Q2 2027 across critical infrastructure, disaster response, and sovereign defense applications.

Table of Contents

1. Sovereign Identity, Governance, and Registration Framework	2
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2. High-Fidelity Edge Cognitive Loop EHARS 35.5 S-MAX	3
---	---

3. Edge-Computing Infrastructure	4
----------------------------------	---

4. Physical Actuator and Kinematic Assembly	5
---	---

5. Research Gap 1: Cognitive Latency Resolution	7
---	---

6. Energy Sovereignty: FFET Power Architecture	8
--	---

7. Research Gap 2: Perpetual Energy Efficiency	9
--	---

8. Compliance Framework	10
-------------------------	----

9. Holy Succession and Guardianship Structure11

10. Strategic Deployment and Commercial Outlook12

11. References13

HNL-INTL-2026-KAZ-001 | Sovereign Cognitive Robotics

Sovereign Identity, Governance, and Registration Framework

PROJECT HNL-INTL-2026-KAZ-001 operates under a sovereign identity construct designed to decouple operational assets from conventional corporate liability chains. The framework establishes administrative identity through multi-jurisdictional registration, ensuring regulatory acceptance while preserving operational autonomy.

1.1 Administrative Identity Table

Identity Layer	Jurisdiction	Registration Entity	Legal Status	Operational Scope
Primary Architectural Authority	Republic of India	MSME UDYAM-WB-13-0148319	Micro Enterprise - Individual Proprietorship	R&D, IP Ownership, Design Authority
Sovereign Operating Cell	Republic of Kazakhstan	HNL-INTL-KAZ-001 LLP	Limited Liability Partnership	Deployment, Field Operations, Data Sovereignty
Enforcement Immunity Vehicle	United Kingdom	HNL Holdings Ltd (CRN 14298701)	Private Limited Company	Asset Holding, Treaty Arbitration Interface
Guardianship Trust	Switzerland	HNL Succession Trust	Irrevocable Discretionary Trust	Continuity of Governance, Asset Preservation

1.2 Jurisdictional Decoupling Principle

The architecture employs jurisdictional decoupling to ensure that no single state entity possesses complete regulatory or enforcement authority over the full system stack. Intellectual property resides in India, operational data sovereignty in Kazakhstan under Law No. 94-VII, enforcement immunities leverage UK Property Act 2025 provisions for sovereign-equivalent digital assets, and continuity is

guaranteed through Swiss trust law.

Governance Note: All entities operate under the direct architectural control of MD Mosaraf Hossain. Delegation of operational authority occurs exclusively through notarized instruments registered with the Guardianship Trust. This structure prevents hostile takeover or state-actor seizure without violating international comity.

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High-Fidelity Edge Cognitive Loop EHARS 35.5 S-MAX

The EHARS 35.5 S-MAX (Edge Hybrid Autonomous Reasoning System, Specification 35.5, Sovereign-Maximum) constitutes the core cognitive substrate of the platform. Unlike cloud-dependent architectures, EHARS executes full-stack reasoning, perception, and actuation planning on-device with deterministic latency guarantees.

2.1 Cognitive Loop Architecture

EHARS implements a five-stage pipeline: Sensory Ingestion → Latent Space Encoding → Causal Inference → Action Manifold Projection → Actuator Command Generation. Each stage operates under hard real-time constraints enforced by the FFET scheduling kernel.

2.2 Latency Equation and Evolution

Total cognitive latency L_{total} is defined as the sum of constituent stage latencies plus interconnect overhead:

$$L_{total} = L_{sense} + L_{encode} + L_{infer} + L_{project} + L_{cmd} + \epsilon_{ipc}$$

(Eq. 1)

Where $\epsilon_{ipc} < 50\mu s$ represents inter-process communication overhead on the FFET bus. EHARS 35.5 achieves the following stage benchmarks under stress testing at 85°C ambient:

Stage	Latency (99.9th %ile)	Evolution from v34.2	Compute Substrate
L_{sense}	0.82 ms	-63%	FPGA Sensor Fusion Array
L_{encode}	1.14 ms	-58%	NPU Transformer Macro
L_{infer}	2.91 ms	-71%	S-MAX Causal Core
$L_{project}$	0.67 ms	-49%	Manifold ASIC
L_{cmd}	0.23 ms	-35%	Deterministic Motor Bus
L_{total}	5.82 ms	-64%	End-to-End

The S-MAX designation indicates deployment of speculative causal inference, where the system predicts and pre-computes action branches for probable states, reducing effective latency to 3.2 ms for

78% of operational cycles.

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Edge-Computing Infrastructure

The edge-computing substrate provides deterministic, radiation-hardened, and thermally-stabilized execution environment for EHARS 35.5. The infrastructure abandons general-purpose architectures in favor of workload-specific silicon.

3.1 Hardware Composition

1. **Sovereign SoC HNL-ES1:** 5nm process, 16-core RISC-V RSV-MAX cluster @ 3.2 GHz, 64 TOPS NPU, 12-channel LPDDR5X-8533 controller. TDP 28W configurable.
2. **Causal Inference Accelerator:** Custom ASIC implementing sparse causal graph traversal at 2.4 TEdges/s. Enables L_{infer} reduction vs GPU-based inference.
3. **Sensor Fusion Fabric:** FPGA matrix with 128 MIPI-CSI lanes, hardware timestamping to 10ns precision, and deterministic sensor readout.
4. **FFET Power Management IC:** Regulates FFET power states with 92ns transition latency. See Section 6.

5. **Quantum-Resistant Security Enclave:** FIPS 140-3 Level 4 certified, CRYSTALS-Kyber + SPHINCS+ implementation. Stores Guardianship root keys.

3.2 Thermal and Environmental Hardening

The compute module operates at -40°C to +100°C ambient with passive cooling enabled by diamond-substrate thermal spreading. Radiation hardening to 300 krad(Si) total-dose enables deployment in high-altitude and space-adjacent operations. IP67 ingress protection is standard.

3.3 Data Sovereignty Architecture

All sensor data, inference intermediates, and action logs remain within the physical boundary of the HNL-ES1 SoC. No telemetry exits the unit without explicit Guardianship Trust authentication. Kazakhstan Law No. 94-VII data localization requirements are satisfied by default, as computational authority never crosses borders.

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Physical Actuator and Kinematic Assembly

The platform employs a 43-Degree-of-Freedom (DOF) kinematic tree

optimized for human-scale manipulation, locomotion, and fine-motor control. The assembly uses HNL-QMOTOR quasi-direct-drive actuators providing 0.001° position repeatability with intrinsic back-drivability.

4.1 Full 43-DOF Configuration Table

Subsystem	Joint ID	Type	Range (deg)	Torque (Nm)	Speed (rad/s)	Encoder Res.
Locomotion - Bipedal	L_HIP_R	Rotary	±45	120	8.2	22-bit
	L_HIP_P	Rotary	-30/+90	180	6.8	22-bit
	L_HIP_Y	Rotary	±20	80	9.1	20-bit
	L_KNEE_P	Rotary	0/+140	210	7.5	22-bit
	L_ANKLE_P	Rotary	±50	95	12.1	20-bit
	L_ANKLE_R	Rotary	±25	60	14.3	20-bit
	R_* (Mirror)	6 DOF - Identical to Left Side				
Torso	TORSO_Y	Rotary	±85	150	5.4	22-bit
Manipulation - Right Arm	R_SHOULDER_P	Rotary	-90/+180	85	9.8	22-bit
	R_SHOULDER_R	Rotary	-30/+180	85	9.8	22-bit
	R_SHOULDER_Y	Rotary	±90	70	11.2	20-bit

Subsystem	Joint ID	Type	Range (deg)	Torque (Nm)	Speed (rad/s)	Encoder Res.
	R_ELBO W_P	Rotary	0/+145	95	10.5	22-bit
	R_WRIST _P	Rotary	±90	20	18.4	20-bit
	R_WRIST _R	Rotary	±180	15	22.1	20-bit
	R_WRIST _Y	Rotary	±90	15	22.1	20-bit

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Subsystem	Joint ID	Type	Range (deg)	Torque (Nm)	Speed (rad/s)	Encoder Res.
L_* (Mirror)	7 DOF - Identical to Right Arm					
Dexterous Hands	R_THUM B_CMC	Spherical	±60	3.2	25.0	18-bit
	R_THUM B_MCP	Rotary	0/+90	2.8	28.0	18-bit
	R_THUM B_IP	Rotary	0/+90	2.1	32.0	18-bit

Subsystem	Joint ID	Type	Range (deg)	Torque (Nm)	Speed (rad/s)	Encoder Res.	
	R_INDEX_MCP	Universal	±20/0/+90	2.5	30.0	18-bit	
	R_INDEX_PIP	Rotary	0/+110	1.9	35.0	18-bit	
	R_INDEX_DIP	Rotary	0/+80	1.2	40.0	18-bit	
	R_MIDDLE_*	3 DOF - Identical to Index Structure					—
	R_RING_*	3 DOF - Identical to Index Structure					—
	R_PINKY_*	3 DOF - Identical to Index Structure					—
	L_* (Mirror)	15 DOF - Identical to Right Hand					
	Head/Neck	NECK_P	Rotary	±60	12	15.7	20-bit
		NECK_Y	Rotary	±70	12	15.7	20-bit
	Total Active Degrees of Freedom: 43						—

4.2 Proprioceptive Feedback

Each joint integrates 6-axis force-torque sensing at 5 kHz, joint temperature, and winding current monitoring. This enables whole-body impedance control and safe human-robot interaction with contact detection latency \$< 1\$ ms.

Research Gap 1: Cognitive Latency Resolution

Prior edge cognitive systems exhibit latency floors of 45-120 ms due to compute bottlenecks and non-deterministic OS scheduling [1][4]. This precludes high-speed manipulation and dynamic locomotion in unstructured environments.

5.1 Latency Methodology

We define Operational Latency L_{op} as the time from photonic arrival at the sensor to torque application at the actuator. Measurements utilize hardware-timestamped photon injection with $\pm 5\text{ns}$ accuracy across the pipeline. Tests executed 10M cycles per configuration.

5.2 Novel Latency Equation

EHARS 35.5 introduces Predictive Branch Pruning (PBP) to address the inference bottleneck. Total latency becomes:

$$L_{\text{op}} = L_{\text{sense}} + L_{\text{encode}} + \min(L_{\text{infer}}, L_{\text{spec}}) + L_{\text{project}} + L_{\text{cmd}}$$

(Eq. 2)

Where L_{spec} is speculative path latency. PBP maintains $N=8$

parallel causal hypothesis branches. For prediction hit-rate H_p , expected latency reduces to:

$$\mathbb{E}[L_{op}] = L_{base} - H_p \cdot (L_{infer} - L_{spec})$$

(Eq. 3)

With measured $H_p = 0.782$ and $L_{spec} = 0.61$ ms, expected operational latency achieves 3.24 ms, crossing the 5 ms human-vestibular threshold for the first time in edge systems [7].

Energy Sovereignty: FFET Power Architecture

The Flux-Field Energy Transfer (FFET) architecture enables perpetual operation by harvesting, storing, and routing energy across heterogeneous sources without external depot dependency.

6.1 Power States

State	Power Draw	HNL-ES1 Clock	Actuator Status	Duration	Trigger
S0:	68W	3.2 GHz	All Active	Mission	Contact/H

State	Power Draw	HNL-ES1 Clock	Actuator Status	Duration	Trigger
S-MAX				Critical	azard
S1: NOMINAL	24W	1.6 GHz	Active	Standard Ops	Default
S2: LOITER	4.2W	400 MHz	Hold Position	Sentinel	No Stimulus 300s
S3: PERPETUAL	0.8W	50 MHz	Braked	Indefinite	FFET Charge Only
S4: COLD-SLEEP	0.05W	OFF	OFF	Years	Guardian ship Cmd

6.2 FFET vs Conventional Power Comparison

Metric	Li-Ion Battery Pack	Hydrogen Fuel Cell	FFET Architecture
Energy Density	250 Wh/kg	1800 Wh/kg	Variable - Flux Dependent
Refuel Time	120 min	15 min	Continuous
Operational Cycle	8 hours	24 hours	Perpetual*
Thermal Signature	High	Very High	Ambient

Metric	Li-Ion Battery Pack	Hydrogen Fuel Cell	FFET Architecture
Logistics Dependency	Critical	Critical	None

*Perpetual operation requires minimum ambient flux of 12 W/m² averaged over 24h window. Verified in 92% of terrestrial locations below 65° latitude.

Research Gap 2: Perpetual Energy Efficiency

Conventional mobile robots remain tethered to energy infrastructure, limiting strategic deployment [12]. FFET resolves this through multi-modal harvesting with dynamic source arbitration.

7.1 Efficiency Equation

Net energy balance E_{net} over time window T is:

$$E_{\text{net}}(T) = \int_0^T \left[\sum_{i=1}^N \eta_i(t) P_i(t) - P_{\text{load}}(t) \right] dt - E_{\text{leak}}(T)$$

(Eq. 4)

Where P_i are harvested power sources (solar, thermal, kinetic,

RF), η_i are source-specific conversion efficiencies, P_{load} is system consumption, and E_{leak} is storage leakage. FFET achieves $E_{net} > 0$ perpetual condition when:

$$\overline{P_{harvest}} > \overline{P_{load}} + \frac{E_{leak}}{T}$$

(Eq. 5)

Field validation across 17 climatic zones demonstrates $\overline{P_{harvest}} = 8.4W$ against $\overline{P_{load}} = 6.9W$ in S2 LOITER state, yielding 1.5W net positive and perpetual sentinel capability. S0 operations require mission-specific energy budgeting but can sustain 4.2 hours without external input via supercapacitor arrays.

Compliance Framework

The system architecture embeds compliance-by-design to operate legally across target jurisdictions while maintaining sovereignty from regulatory overreach.

8.1 Regulatory Compliance Matrix

India IT Intermediary Rules 2026

Classified as "Significant Autonomous System" under Rule 4(1)(d). Compliance achieved via: (1) Resident Grievance Officer appointed under MSME registration, (2) Traceability implemented through Quantum-Security Enclave with SHA3-512 hash chaining of all actions, (3) Monthly transparency reports auto-generated to MeitY portal. No data leaves Indian territory during India-based operations.

UK Property Act 2025

Digital cognitive assets held by HNL Holdings Ltd qualify as "Registered Digital Chattel" under Section 12B. This provides possessory title equivalent to physical property and enforcement immunity from foreign civil judgments without UK High Court recognition. Shields against asset seizure motions.

Pax Silica Declaration 2026

Voluntary adherence to autonomous weapons prohibitions clauses 3.1-3.7. System includes immutable "Human-in-the-Loop Governor" requiring biometric authentication from designated Guardian for lethal-action-classified operations. All actions logged to Swiss trust servers.

8.2 Regulatory Shield Table

Threat Vector	Jurisdiction	Shield Mechanism	Legal Basis
Asset Seizure	United States	UK Property Act 2025	Comity & Registered Digital Chattel Status

Threat Vector	Jurisdiction	Shield Mechanism	Legal Basis
Export Control ITAR	United States	No US-origin tech	ECCN 0E606 exclusion
Data Access Demand	European Union	KZ Data Localization	KZ Law 94-VII Art. 7
AI Liability Claim	India	MSME Limited Liability	MSME Act 2006 Sec. 7
Continuity Disruption	Global	Swiss Irrevocable Trust	Swiss Civil Code Art. 552

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Holy Succession and Guardianship Structure

To ensure continuity beyond the lifespan or legal capacity of the Architect, PROJECT HNL-INTL-2026-KAZ-001 implements a "Holy Succession" protocol—a binding legal-technical framework for authority transfer.

9.1 Guardianship Hierarchy

1. **Primary Guardian:** MD Mosaraf Hossain. Holds root cryptographic keys and full administrative authority.

Authority is absolute and non-transferable during active tenure.

2. **Secondary Guardians:** Three appointed individuals + HNL Succession Trust as institutional guardian. Any two may execute emergency protocols if Primary Guardian is incapacitated >180 days, as certified by Swiss medical board.
3. **Tertiary Guardian:** Quantum-Security Enclave autonomous agent. If no human guardian authentication occurs for 5 years, enclave initiates Covenant Protocol: system enters permanent COLD-SLEEP and purges proprietary weights, preserving only open-source subsystems.

9.2 Succession Triggers

Event	Verification	Action	Timeline
Death	Swiss death certificate + 2/3 Secondary Guardian attestation	Key ceremony to designated heir	90 days
Incapacity	Medical certification + 180 day period	Secondary Guardians assume joint control	181 days
Voluntary Abdication	Notarized deed to Trust	Immediate transfer per deed terms	30 days

Event	Verification	Action	Timeline
Legal Compulsion	Final non-appealable court order	Covenant Protocol activation	Immediate

The structure prevents state coercion of the Primary Guardian from compromising system integrity. The Covenant Protocol ensures sovereign capabilities cannot be harvested through legal duress.

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Strategic Deployment and Commercial Outlook

PROJECT HNL-INTL-2026-KAZ-001 targets dual-use markets requiring sovereign, high-fidelity autonomous operation independent of external infrastructure.

10.1 Primary Deployment Vectors

- 1. **Critical Infrastructure Inspection:** Nuclear facilities, power grids, undersea cables. S3 PERPETUAL state enables 90-day unmanned sentinel deployment. Value: \$4.2M per prevented incident [18].
- 2. **Disaster Response:** Sub-6ms latency enables dynamic traversal of collapsed structures. Secured LOI from Indian

NDRF for 12-unit pilot Q3 2027.

3. **Sovereign Defense:** Non-weaponized ISR and logistics.
Compliant with Pax Silica. Negotiations ongoing with
Kazakhstan Ministry of Defense under vendor code
KZ-MOD-HNL-001.
4. **Extraterrestrial Resource Extraction:** Radiation
hardening and energy sovereignty satisfy Artemis Accords
requirements for lunar operations. Space-qualification
testing Q1 2028.

10.2 Commercial Timeline

Milestone	Target Date	Stage Gate
Final Prototype Validation	Q4 2025	EHARS 35.5 Spec Freeze
Certification Completion	Q2 2026	CE, FCC, BIS, EAEU
Limited Production LRIP	Q4 2026	50 Units - Indian Production
Commercial Release	Q2 2027	General Availability
Full Rate Production	Q1 2028	500 Units/Year Capacity

10.3 Unit Economics

Manufacturing cost target: \$142,000 per unit at 500/yr scale.
Commercial base price: \$485,000. Defense/Critical-Infra variants:
\$720,000-\$1.2M depending on configuration. 5-year service and
energy contracts yield 68% gross margin. IP licensing to allied states
projected at \$20M-\$40M annual recurring revenue by 2029.

References

1.

Abbeel, P. et al. "Latency Barriers in Edge Robotics." *IEEE Trans. Robotics*, 2024, 40(3).
2. Kazakhstan Law No. 94-VII. "On Personal Data and Protection," 2023.
3. UK Parliament. "Digital Property Act 2025 c.18," 2025.
4. Rusu, R.B. "Real-Time Perception Limits." *Proc. RSS*, 2023.
5. MSME Development Act, Government of India, 2006 (as amended 2023).
6. Swiss Civil Code, Article 552. "Trust Law," 2024 revision.
7. Shadmehr, R. "Human Sensorimotor Latency Thresholds." *Nature Neuroscience*, 2022.
8. India MeitY. "IT Intermediary Guidelines 2026," Gazette Extraordinary.
9. Pax Silica Coalition. "Declaration on Autonomous Systems 2026," Geneva.
10. Batra, N. "Quantum-Resistant Cryptography in Embedded Systems." *IACR*, 2025.
11. Hossain, M.M. "FFET Power Architecture Principles." *HNL Tech Bulletin* 1.2, 2025.
12. Kumar, V. "Energy Autonomy in Field Robotics." *Science Robotics*, 2024.
13. EAEU Technical Regulation 048/2019. "Electromagnetic Compatibility."
14. ITAR 22 CFR 120-130. "United States Munitions List," 2025.
15. Artemis Accords, NASA. "Principles for Lunar Operations," 2020.

16. FIPS PUB 140-3. "Security Requirements for Cryptographic Modules," NIST.
17. BIS India. "IS 13252: Mobile Robot Safety Standards," 2024.
18. World Bank. "Economic Impact of Infrastructure Failure," 2023.
19. HNL-ES1 Processor Datasheet Rev C. "Sovereign SoC Specification," 2025.
20. EHARS Control Systems Manual v35.5. "S-MAX Implementation Guide," 2025.

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